

HINTS & SOLUTION

Q.1 (A)

Q.2 (C)

Q.3 (C)

Q.4 (A)

$$d_{\text{unit cell}} = d_{\text{atom}} \times \text{PF}$$

$$d_{\text{atom}} = \frac{1}{0.1} = 10 \text{ gm/cm}^3$$

$$\text{Volume atom} = \frac{4}{3} \pi (1 \times 10^{-8})^3 \text{ cm}^3$$

$$\text{Mass of atom} = \left(\frac{4}{3} \pi \times 10^{-24} \times 10 \right) \text{ gm}$$

$$\begin{aligned} \text{Atomic mass of element} &= \frac{4}{3} \pi \times 10^{-23} \times 6 \times 10^{23} \\ &= 8\pi \end{aligned}$$

Q.5 (C)

Q.6 (B)

Length of body diagonal $\sqrt{3} a$

Length of body diagonal occupied

$$= r_C + 2r_A + 2r_B + 2r_A + r_C$$

$$= 2r_C + 4r_A + 2r_B$$

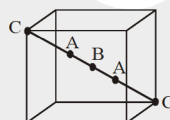
$$= 2r_C + 4 \times 0.225 r_C + 2 \times 0.414 r_C$$

$$= 3.728 r_C$$

$$\text{Also } \sqrt{2}a = 4r_C$$

$$\text{Fraction occupied} = \frac{3.728 r_C}{\sqrt{3} \times \frac{4r_C}{\sqrt{2}}} = 0.7609$$

$$\text{Fraction of body diagonal not covered} = 0.24$$



Q.7 (D)

$A^{2+}(\text{aq})$

$B^-(\text{aq})$

AB_2

So It will be face centered cubic with A^{2+} ions at face centres and corners and B^- occupying all tetrahedral voids.

Q.8 (A)

Q.9 (B)

Q.10 (A)

Q.11 (D)

Q.12 (B)

Q.13 (C)

Q.14 (A) (B) (D)

(a) Frenkel defects are known as dislocation defects

(b) When Al^{3+} is inserted in NaCl 2 cation vacancies are created.

(c) Addition of As in Si results in formation of n-type semiconductors.

(d) Due to vacancy defect packing fraction decreases.

Q.15 (A) (C) (D)

Q.16 (B) (C)

Q.17 (B) (C) (D)

Doping as by Si yield 'n' type semiconductor

Anion can not occupy interstitial voids as anion is bigger in size.

Q.18 (A) (C)

Q.19 (A) P, S (B) P, Q (C) Q (D) Q, R

Q.20 (A) P (B) Q, S (C) R

Q.21 0.25%

Q.22 8

Let $\text{Fe}^{2+} = x$ atom

$2 \times x + 3(7-x) = 16$

$x = 5$

$\text{Fe}^{+2} = 5$ atom

and $\text{Fe}^{+3} = 2$ atom

In $2\text{Fe}^{+3} \dots\dots 3\text{Fe}^{+2}$

$\therefore$ 1 cation vacancies in 8 atom

In 8 atom $\dots\dots$ 1 cation vacancies

$\therefore 100 \dots\dots \frac{100}{8} \% \quad \text{Ans.}$

Q.23 FCC

Q.24 4 molecules of H_2O

Q.26 $6.02 \times 10^{18} \text{ mol}^{-1}$

Q.27 4, 6, 8

Q.28 ZnAl_2O_4

Q.29 (a) octahedral, (b) $1/4$

Q.30 0067

$4800 - 15x + 15y = 4795$

$x - y = \frac{1}{3} \quad ; \quad x + y = 1$

$x = \frac{2}{3} \quad y = \frac{1}{3}$

$\% = \frac{2}{3} \times 100 = 66.667$

